

S.N.V. Sai Narasimha Madhav

Naga Venkata Sai Narasimha Madhav Sankurathripati

MS (by Research) Scholar · Medical Science & Technology · IIT Madras

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RESEARCH PROFILE

MS (by Research) scholar in the Computational Neuroscience Lab, IIT Madras, working at the intersection of clinical biomechanics and brain-inspired machine learning. My work builds machine-learning and computer-vision pipelines that turn ordinary clinic video into quantitative motor biomarkers — knee asymmetry, trunk lean, pelvis sway — for the detection and severity classification of gait abnormality in children with cerebral palsy (CP). A single throughline runs across every system I build: **raw movement → pose → signal → decision**, with oscillator-based neural networks (DONN / ResHopf) as a recurring modelling tool.

RESEARCH INTERESTS

Markerless video-based gait analysis · Cerebral palsy and pediatric rehabilitation · Computational neuroscience and motor control · Spatio-temporal graph neural networks · Oscillatory and dynamical-systems models of movement · Bio-signal analysis (HD-EMG) · Reinforcement learning for musculoskeletal locomotion · Translational clinical software.

EDUCATION

DEGREE / EXAMINATION	INSTITUTION / BOARD	YEAR	RESULT
M.S. by Research — Medical Science & Technology	Indian Institute of Technology Madras	2024 – present	CGPA 8.0
B.Tech. — Electronics & Communication Engineering	IIITDM Kancheepuram	2020 – 2024	CGPA 8.13
Intermediate (Class XII)	Board of Intermediate Education, Andhra Pradesh	2020	74.0%
Secondary (Class X)	Central Board of Secondary Education	2018	76.4%

RESEARCH EXPERIENCE

Research Scholar — Computational Neuroscience Lab, Dept. of Biotechnology, IIT Madras 2024 – present

Advisor: Prof. V. Srinivas Chakravarthy

- Building an end-to-end markerless gait-analysis pipeline for cerebral palsy: BlazePose (MediaPipe) 2D pose estimation from frontal- and sagittal-plane video, joint-angle and asymmetry signal extraction, gait-cycle segmentation, and downstream classification.
- Designing and validating discriminative biomechanical features (knee-angle asymmetry, trunk lean, pelvic obliquity and sway) as interpretable motor biomarkers that clinicians can read directly.
- Developing oscillator-based classifiers (DONN / ResHopf, built on Hopf-oscillator dynamics) and spatio-temporal graph convolutional networks (ST-GCN) for CP-vs-typical discrimination and GMFCS severity grading.
- Applying operator-theoretic methods (Koopman / Hankel-DMD) to gait time series to obtain linear, interpretable descriptions of nonlinear gait dynamics.
- Delivering a clinician-facing web platform for CP gait assessment with longitudinal patient tracking.

Undergraduate Research — Deep Learning for Medical Imaging 2022

Mentor: Dr. Priyanka Kokil, Associate Professor, Dept. of ECE, IIITDM Kancheepuram

- Critical review of deep-learning literature across diabetic retinopathy grading, skin-cancer classification, vision transformers, and cerebral palsy — the entry point to my current CP research direction.

Research Intern — Plastic Degradation Analysis 2022

Mentor: Dr. Madan Kumar Lakshmanan, Principal Scientist, CSIR-CEERI

- Applied deep-learning techniques to near-infrared (NIR) spectroscopy data for characterising plastic degradation.

Digital Design Project — Hardware Sorting Architecture 2022

Mentor: Dr. Binsu J. Kailath, Professor, Dept. of ECE, IIITDM Kancheepuram

- Implemented a sorting algorithm in Verilog HDL; carried out simulation, synthesis and implementation in Xilinx Vivado.

SELECTED PROJECTS

Pediatric CP Gait Clinic — clinical web platform

FastAPI · React · MediaPipe · SQLite · PyTorch

- Full clinical web application for CP gait assessment: patient management across the GMFCS, MACS, CFCS and EDACS classification systems; session-based video upload; frontal and sagittal analysis with threshold-based metric cards; longitudinal tracking across visits.

Frontal-Plane Gait Classification — ST-GCN

ST-GCN · BlazePose · PyTorch

- CP-vs-typical classification from BlazePose keypoints, modelling the body as a spatio-temporal graph of joints evolving over the gait cycle.

DONN · ResHopf Classifier — physics-grounded deep learning

Hopf oscillators · Dynamical systems · PyTorch

- An oscillatory deep network built on Hopf-oscillator dynamics, reused as a common backbone across gait, surface EMG and upper-limb kinematics.

DCP Reaching Classification — open-source benchmark

Tri-branch ST-GCN / CNN / Hopf · KU Leuven DCP dataset (Vanmechelen et al., 2024)

- Classification of dyskinetic CP versus typically-developing upper-limb reaching.
- Reported honestly: window-level data leakage inflated the initial scores; the corrective protocol is subject-wise cross-validation.

CP Locomotion RL — musculoskeletal simulation

MyoSuite · MuJoCo · PPO · SAC

- Musculoskeletal locomotion learning with PPO and SAC, a curriculum scheduler, a CP-impairment injection model, and a stability gate on the learned policy.

GMFCS Severity Classifier

DONN · OpenPose · Stanford mobile-gaitlab dataset (Kidziński et al., 2020)

- DONN-based GMFCS severity grading from clinical gait video, with automatic camera-view correction.

HD-EMG Intent Classification

HD-EMG · DONN-ResHopf · PyTorch

- Classification of elbow isometric contractions from a high-density surface-EMG dataset (Mendeley Data).

TECHNICAL SKILLS

Languages	Python, C, Verilog HDL
ML & Deep Learning	PyTorch, ST-GCN, graph neural networks, CNNs, oscillatory networks (DONN / ResHopf), reinforcement learning (PPO, SAC)
Computer Vision	MediaPipe BlazePose, OpenPose, markerless pose estimation, video pre-processing and view correction
Simulation	MuJoCo, MyoSuite, Xilinx Vivado
Web & Data	FastAPI, React, SQLite
Domain Expertise	Gait biomechanics, clinical movement analysis, image processing, pattern recognition, bio-signal analysis, dynamical-systems and operator-theoretic methods

CERTIFICATIONS

- Introduction to Neuroscience for Engineers — NPTEL, Govt. of India (2022)
- Python Bootcamp — Udemy (2022)
- Python 3: From Beginner to Pro — Udemy (2021)
- Internet of Things Workshop — LEMA Labs, IIT Madras (2021)
- Internet of Things Master Class — Pantech E-Learning (2021)
- KLA Workshop on AI and HPC in Semiconductor Manufacturing (2021)

AWARDS & ACHIEVEMENTS

- Scored 160/160 in the A1 German language proficiency examination (2022)
- Second Prize, National Science Olympiad (2018)
- Third Prize, International English Olympiad (2018)
- Second Prize, Innovation Ideas Competition, CSIR-CLRI (2015)

POSITIONS OF RESPONSIBILITY

- Placement Volunteer, IIITDM Kancheepuram (2022 – 2024)
- Coordinator, Quiz Club (2021 – 2024)
- Class Representative (2020 – 2024)
- Student Volunteer, SAMGATHA — institute cultural festival

LANGUAGES & INTERESTS

Languages	English (read, write, speak — proficient) · Telugu (native) · Tamil · Hindi (read, write) · Sanskrit (read, write) · German (A1, elementary)
Interests	Chess (architected a full chess platform) · music production · batting-biomechanics analysis in cricket · writing poetry · KP and Vedic astrological computation with real ephemeris data

S.N.V. Sai Narasimha Madhav · Chennai, India